



Pediatric Tachycardia

Wide Complex (> 0.09 sec)

History

- Past medical history
- Medications or Toxic Ingestion (Aminophylline, Diet pills, Thyroid supplements, Decongestants, Digoxin)
- Drugs (nicotine, cocaine)
- Congenital Heart Disease
- Respiratory Distress
- Syncope or Near Syncope

Signs and Symptoms

- Heart Rate: Child > 180/bpm
Infant > 220/bpm
- Pale or Cyanosis
- Diaphoresis
- Tachypnea
- Vomiting
- Hypotension
- Altered Level of Consciousness
- Pulmonary Congestion
- Syncope

Differential

- Heart disease (Congenital)
- Hypothermia/ Hyperthermia
- Hypovolemia or Anemia
- Electrolyte imbalance
- Anxiety/ Pain/ Emotional stress
- Fever/ Infection/ Sepsis
- Hypoxia, Hypoglycemia
- Medication/ Toxin/ Drugs (see HX)
- Pulmonary embolus
- Trauma, Tension Pneumothorax

Assess tachycardia in context of clinical condition
Identify and treat underlying cause of tachycardia

Unstable / Serious Signs and Symptoms
AMS, shock, hypotension
HR Typically > 180 Child
HR Typically > 220 Infant

YES → P

NO

B	12 Lead ECG Procedure
P	Cardiac Monitor
	IV or IO Access Protocol UP 6

Cardiac Monitor

Cardioversion Procedure

0.5 – 1 J / kg
Repeat and increase to **2 J / Kg**

Consider sedation
Do NOT delay cardioversion
Midazolam 0.1 – 0.2 mg / kg IV / IO / IN

May repeat if needed
Maximum Single Dose 2 mg
Maximum Total Dose 5 mg

**Single lead ECG able to
diagnose and treat arrhythmia**

12 Lead ECG not necessary to diagnose and
treat, but preferred when patient is stable.

Regular Rhythm?

NO →



Contact Medical Control



YES

Probable Ventricular Tachycardia

If QRS Regular and Monomorphic
Adenosine 0.1 mg / kg IV / IO
Maximum 6 mg

May repeat
Adenosine 0.2 mg / kg IV / IO
Maximum 12 mg

Expert consultation recommended
for further medication management
and/ or cardioversion procedure



**Notify Destination or
Contact Medical Control**



AT ANY TIME

Pulseless



**Go to
Pediatric Pulseless
Arrest Protocol**



Pediatric Tachycardia

Wide Complex (> 0.09 sec)

Pearls

- **Recommended Exam:** Mental Status, Skin, Neck, Lung, Heart, Abdomen, Neuro
- **Monomorphic QRS:**
All QRS complexes in a single lead are similar in shape.
- **Polymorphic QRS:**
QRS complexes in a single lead will change from complex to complex.
- Use length-based or weight-based pediatric resuscitation system for medication, equipment, cardioversion, and defibrillation guidance. Pediatric paddles should be used in children < 10 kg.
- Rhythm should be interpreted in the context of symptoms and pharmacological or electrical treatment given only when symptomatic, otherwise monitor and reassess.
- **12-Lead ECG:**
12-Lead ECG is not necessary to diagnose and treat arrhythmia. A single lead ECG is often all that is needed.
Obtain 12-Lead when patient is stable and/ or following a rhythm conversion.
When administering adenosine, obtaining a continuous 12-Lead can be helpful later to physicians.
- **Unstable condition:**
Condition which acutely impairs vital organ function and cardiac arrest may be imminent.
If at any point patient becomes unstable move to unstable arm in algorithm
- Document all rhythm changes with monitor strips and obtain monitor strips with each therapeutic intervention.
- **Serious Signs and Symptoms:**
Respiratory distress/ failure.
Signs of shock/ poor perfusion with or without hypotension.
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Sudden collapse with rapid, weak pulse
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- **Wide Complex Tachycardia (≥ 0.09 seconds):**
SVT with aberrancy.
VT: Uncommon in children. Rates may vary from near normal to > 200 / minute.
Most children with VT have underlying heart disease / cardiac surgery/ long QT syndrome/ cardiomyopathy.
Amiodarone 5 mg / kg over 20 – 60 minutes or Procainamide 15 mg / kg over 30 – 60 minutes IV / IO are recommended agents. They should not be administered together. Consultation with Medical Control is advised when these agents are considered.
- **Torsade's de Pointes/ Polymorphic (multiple shaped) Tachycardia:**
Rate is typically 150 to 250 beats/ minute.
Associated with long QT syndrome, hypomagnesaemia, hypokalemia, many cardiac drugs.
May quickly deteriorate to VT.
Separating the child from the caregiver may worsen the child's clinical condition.
- Monitor for respiratory depression and hypotension associated if Diazepam, Lorazepam, or Midazolam is used.
- Continuous pulse oximetry is required for all SVT patients if available.